

MAEPiMS Challenge 2025

COVID-19 Prediction Challenge for Nigeria: 2020-2022

This document outlines the framework for a collaborative challenge to predict COVID-19 daily cases in Nigeria, utilizing data sourced from Worldometer and other associated repositories. This initiative aims to support public health planning and intervention strategies in response to the COVID-19 pandemic by providing accurate predictions of daily cases from 2020 to 2022 and other future emerging and reemerging respiratory diseases. Interested participants are encouraged to leverage the resources provided for parameterization or as initial values.

NB: THE DEVELOPED MODEL MUST BE ORIGINAL AND NOT DERIVED FROM EXISTING LITERATURE. PARTICIPANTS ARE ENCOURAGED TO INCORPORATE NEW IDEAS AND INNOVATIVE APPROACHES THAT GO BEYOND ESTABLISHED FRAMEWORKS.

Overview of the Challenge

COVID-19 has posed significant health and socioeconomic challenges globally, with Nigeria experiencing unique impacts due to its demographic, economic, and infrastructural context. Accurate nowcasts and forecasts of COVID-19 daily cases can provide actionable insights for public health officials, policymakers, and other stakeholders to mitigate the effects of the pandemic and allocate resources effectively.

Challenge Details

- Objective: To predict daily COVID-19 cases in Nigeria for the period spanning 2020 to 2022. Fit the data for 2020-2021 and forecast for 2022 using a mathematical model. The best fit will be of interest.
- Data Sources:
 - Primary data: [Worldometer COVID-19 daily cases for Nigeria](#)
 - Supplementary demographic, economic, and health data:
 - ✓ [Nigeria Demographics](#)
 - ✓ [Nigeria Centre for Disease Control \(NCDC\) Sitrep Reports](#)
 - ✓ [National Bureau of Statistics Data Portal](#)
 - ✓ [Central Bank of Nigeria Rates](#)
 - ✓ [Our World in Data for Nigeria](#)
 - ✓ [World Health Organization \(WHO\) Data for Nigeria](#)

Prediction Targets Participating teams will provide predictions for the following targets:

1. Primary Target: Daily confirmed COVID-19 cases for Nigeria during the 2020-2022 period. Predictions should include nowcasts and forecasts for daily cases spanning a four-week horizon.
2. Additional Targets:
 - ✓ Category probability predictions for the direction and magnitude of changes in daily case counts.
 - ✓ Probability predictions of peak weeks for COVID-19 cases.
 - ✓ Quantile predictions for the peak incidence of daily cases during the specified period.

Submission Guidelines

- **Duration:** Saturday, 8th March (12 noon) – Monday, 10th March 2025 (12 noon).
- **Submission:** Participating teams are required to submit their results by 10th March 2025; (12 noon) (Nigerian Time).
- Submissions should align with epidemiological weeks (Sunday to Saturday).
- **Prediction Formats:**
 - ✓ Quantile predictions for daily COVID-19 cases.
 - ✓ Sample trajectories: Teams may submit temporally connected samples (100 samples per task).
 - ✓ Rate-trend categories: Predictions for categories such as "large increase," "increase," "stable," "decrease," or "large decrease."

Data Evaluation

Forecasts will be evaluated based on COVID-19 daily case data reported by Worldometer and other supplementary datasets. Predictions will be assessed for accuracy and reliability, with a focus on:

- Quantile Forecasts: Alignment with observed daily cases.
- Rate-Trend Categories: Probabilities for changes in case trajectories.
- Seasonal Targets: Accuracy of peak week and peak incidence forecasts.

Evaluation Period

- Historical Data Analysis: Analysis of predictions against reported data from 2020 to 2022.
- Temporal Scope: The evaluation will cover the entire challenge period, including hindcasts for the preceding weeks.

Useful Tools and References

- Tools for converting dates to epidemiological weeks and vice versa:
 - ✓ R packages: MMWRweek, lubridate.
 - ✓ Python packages: pymmwr, epiweeks.
- Thresholds for rate-trend categories will be determined based on historical distributions observed in Nigerian COVID-19 data.

Document Presentation

- Your solution must be submitted in SINGLE PDF (size 15MB)
- The **SINGLE** PDF should contain tables, codes, charts, and graphs.
- Ensure the preparation of the solution to be typed and submitted is within the allocated time as more time will not be given. At the expected time, the portal closes.
- What is expected in your solution **SINGLE PDF**:
 - **Title**
 - **Justification of your hypothesis and assumptions**
 - **Model developed**
 - **Results and Discussions**

- **Recommendation**
- **References**

Contact Information

For inquiries regarding the COVID-19 Prediction Challenge for Nigeria, please contact: Project Coordinator: maepims@adejimiadeniji.com

Participants are encouraged to use the links provided above to gather data, derive initial values, and parameterize their models. This collaborative effort aims to enhance understanding of COVID-19 dynamics in Nigeria and inform data-driven decision-making to mitigate the pandemic's impact.